

Data Centers & AI: Business Concept

Thesis

Data centers are absorbing an unprecedented, historically unusual scale of capital: over \$1 trillion in global capex in 2026 alone, on a trajectory toward roughly \$7.6 trillion cumulative spend by 2031. That capital, along with the physical build-out it funds, is geographically lopsided: something like 75% of the data center capacity currently under construction globally is going up in the US (BloombergNEF).

The Objective

The goal is to connect the global demand for AI with investment dollars looking for exposure to this category by creating an interactive map where users earn points/tokens for contributing data and are charged points/tokens for accessing the data.

Why now: Over \$1 trillion is being spent building data centers globally this year, driven by AI. About three-quarters of that is happening in the US, but the rest of the world is going to need this infrastructure too, and almost nobody is tracking those other markets.

The Product

What it is: A gated website that maps data centers around the world — where they are, who owns them, whether they can actually get electricity, and whether someone can invest in them. Right now nobody puts all of that in one place. The existing tools either just show pins on a map for free, or charge a lot of money to big companies for research reports — none of them show ownership, deal status, or a way to actually invest in a specific project.

The project starts with a map, similar in spirit to public oil-tanker tracking maps, though more detailed — closer to what a site like Baxtel's map attempts, but with far more depth.

What's on the map:

- Data centers — existing and planned, with basics like size, owner, and status (under construction, operating, etc.)
- Power grid status — whether a given project can actually get electricity, which is the hardest and most valuable thing to track, since a lot of “planned” data centers are stuck for years waiting on a grid connection
- Who owns it, and whether it's for sale — public company, private company, open to investment, recently sold, and to/from whom
- The stuff that builds up around a big campus — workforce housing, hotels, staging yards — as an early signal a project is really happening
- A demand layer — where there's a lot of population and money that would actually use computing power, versus where there isn't, so the map can show “data deserts” (places that need more data centers but don't have them)

The crypto component: let anyone submit updates (a new project, an ownership change, a grid-status update) and get paid in crypto for it — which is a cheaper way to keep a global map accurate than hiring researchers everywhere. Charge in tokens to access the data.

Later, people would also be able to actually invest in specific projects through the site, with crypto handling the payment.

Current Gaps

Where the gaps are: downstream from the data center itself.

The “Shadow Map” idea (construction-worker ripple around a campus) is one instance of a bigger pattern: every layer downstream of the data center itself is under-mapped relative to the data center pins. In rough order of how under-served they are:

- Power grid & interconnection status — this is the biggest gap and the biggest number. The US interconnection queue is backlogged to roughly 2,600 GW with median waits near five years (worse for large loads); interconnection costs alone can eat 30–37% of a withdrawn project's budget. Nobody publishes, at the project level, where a site sits in queue, what its interconnection agreement status is, or which specific transmission/substation upgrade is gating it. This is more valuable and harder to get than land data, and it's the layer that actually determines whether a “planned” data center ever gets built.
- On-site / behind-the-meter power — gas turbines, batteries, and the handful of SMR and geothermal deals (Microsoft/Brookfield, Google, the Brookfield–Bloom Energy partnership) are being announced faster than anyone is tracking who owns the generation asset versus who owns the data center it feeds. That's a distinct, mappable financial-exposure layer.
- Transmission & substation build-out — the actual construction projects (new lines, upgraded substations) that unlock a region's data-desert status. These are often public-utility-rate-based or independent-transmission-company projects with their own investable structures (rate base, YieldCos, infrastructure funds) — currently invisible on any data-center map.
- Water & cooling infrastructure — large campuses can use millions of gallons/day; water-rights and cooling-infrastructure disputes are becoming a siting constraint the way power is, and are essentially unmapped.
- Fiber / network backbone builds — new long-haul and metro fiber routes get built alongside campuses (and are sometimes the reason a “data desert” isn't viable even with power); existing connectivity directories cover some of this but don't connect it to ownership or investability.
- Ancillary real estate (the Shadow Map) — workforce housing, hotels, logistics/staging yards around multi-year construction campaigns. Still nobody else's product.
- Land banked / entitled sites with no operator yet — the screening-tool layer described above; genuinely proprietary if we hand-research it, and the fastest-decaying data of anything on this list.
- Emerging-market data deserts — every incumbent above is North America/Europe-weighted. Latin America, parts of Africa, and Southeast Asia are exactly the regions where the map would be thinnest for everyone else and where being first (with real ownership/deal data, not just pins) is most defensible.
- Ownership & transaction history — the layer that's genuinely missing from every competitor above: who owns a given asset today (public co., private co., fund, sovereign entity), whether it's open to investment, whether it's currently listed, and who it was last bought/sold from and to.

Competitive Landscape

Who else maps this, and what they charge. Nobody in this list combines live map + ownership/deal status + financial-exposure path + token economics — that combination is the actual white space, not the map itself. None have ownership/deal-flow-level financial exposure, none pay contributors, and none have a token or stablecoin settlement layer.

Site / product	What it is	Gated?	Approx. cost	Notes
Data Center Map	Ad-sponsored map with data centers worldwide	No	Free	First mover, but a directory — no pricing, no ownership/deal data, no heatmap.
Baxtel	Rental platform for	Freemium	Free browse; advisory	~\$4M revenue, 13

	leasing data center space		& data engagements are quoted deals	staff (Denver). Closest analog to our free layer — decent design, US-heavy, thin on LatAm/EMEA depth.
Cloudscene	Directory + market intel, carrier/IX/cloud connectivity focus	Freemium	Free directory; paid API/intel tiers not publicly listed	Strong on connectivity data, weak on real estate/power/ownership.
datacenterHawk	Real estate market intelligence: leasing comps, pricing, supply/demand	Gated	Enterprise subscription, quote-only; est. \$10–25M ARR business overall	Acquired by Leeds Equity/Simplify Compliance (2022). Analyst-driven, North America/Europe strongest.
DC Byte	Due diligence for real estate industry about data centers	Gated	Enterprise subscription, quote-only (typically five figures/yr per seat/region)	Broadest global coverage claim; the closest thing to a direct competitor for the paid tiers above.
451 Research / S&P Global Market Intelligence	Enterprise tech research incl. data-center vertical	Gated	\$30–75k+/yr typical for enterprise research seats	Broad, deep, expensive — funds already pay for it, which validates willingness to pay.
Synergy Research Group	Market-share reports, cloud/data-center sizing	Gated per report	\$1,000s per report; subscriptions available	Backward-looking market sizing, not a live map or deal tool.
Cushman & Wakefield “Athena”	Internal broker site-selection/analytics tool	Not sold externally	Free to C&W clients as part of brokerage relationship	The main incumbent risk to have in mind — well-funded, but it's a client-retention tool, not a product for sale.
JLL / CBRE data-center research	Published market outlook reports (marketing) + bespoke advisory	Reports free; advisory gated	Advisory engagements typically \$50k+	Publish enough to get cited, sell the deep work.
BroadGroup / Datacloud	Events, publications, market intelligence	Mixed	Conference sponsorship/attendance \$\$\$; some free content	More media company than data company.
Uptime Institute	Certification, advisory, some market data	Gated	Per-engagement, often \$50k+	Reputation/certification moat, not a mapping product.

Blockchain-Native precedents & Roadmap

Project	Core Mechanic	Strategic Relevance
RWA.xyz	RWA Tracking Terminal	Analog for deal-status ambitions
Hivemapper / Helium	DePIN Incentives	Models for decentralized data contribution

Total Addressable Market

Layer	Size / figure	Source & context
Global data-center capex, 2026	\$1T+ for the year	Dell'Oro, raised its 2026 outlook past \$1 trillion as hyperscale AI buildouts accelerated.
Top 9 cloud/AI players' combined capex, 2026	~\$830B	TrendForce; ~79% YoY growth, led by Microsoft, Google, Meta, Amazon.
Cumulative AI infrastructure capex, 2026–2031	~\$7.6T	Goldman Sachs baseline model; \$765B in 2026 growing to \$1.6T/yr by 2031.
New data-center capacity, 2026–2030	~100 GW added, doubling global capacity	JLL 2026 Global Data Center Outlook; 14% CAGR.
Real-estate value creation from that buildout	\$1.2T, needing ~\$870B in new debt financing	JLL — this is the financing gap our exposure/marketplace layer sits inside.
Total data-center market revenue (colo, cloud, hosting)	\$416B (2024) → \$620B+ (2029 est.)	Statista Market Insights, via industry roundup.
US interconnection queue backlog	~2,600 GW active requests; median wait ~5 yrs, data centers up to 12 yrs in places	Industry reporting on FERC/ISO queue data — this is the “data desert” layer, quantified.
Interconnection cost burden on withdrawn projects	30–37% of total project budget	Same reporting — explains why power/grid access, not land, is the real bottleneck to map.
Existing data-intelligence incumbents' revenue (proxy for how small this niche still is)	datacenterHawk ~\$10–25M ARR; Baxtel ~\$4M	vs. the \$1T+/yr of capex those tools sit next to — the intelligence layer is a rounding error on the market it serves, which is the whitespace.

Target Audience

Likely buyers, roughly in order of willingness to pay:

- Infrastructure and PE funds, family offices — the existing paying customer base for datacenterHawk/DC Byte/451; they need deal flow and diligence data, and would be the fastest sell for the paid tiers.
- Hyperscaler and large-operator site-selection teams — currently dependent on brokers' internal tools; an independent, faster-updating, more globally complete source is a genuine upgrade for international scouting, especially in markets those internal tools don't cover well.
- Utilities and grid operators — less a buyer than a data partner/user, but worth cultivating as a source and a distribution channel given how central interconnection status is to the product.
- Sovereign and municipal economic-development bodies — emerging-market governments actively courting data-center investment want visibility into what's happening in comparable markets and want to be discoverable by capital; potential partners as well as customers.
- Crypto-native capital seeking real-world-asset yield — this is a stronger pitch than most RWA tokenization because the underlying growth story (AI compute demand) is front-page news already, rather than something that needs explaining. This audience buys the story fast.
- Retail/prosumer crypto users — thematic, small-ticket exposure to “AI infrastructure” without buying a hyperscaler stock or a REIT is a genuinely new product, not a repackaged one.
- Local brokers, landowners, and engineers in underserved markets — not buyers so much as the contributor base the token economy is built to recruit; paying them for verified data is what lets us cover Latin America, Africa, and Southeast Asia faster than any incumbent's in-house analyst team could.

Revenue Estimates

These are illustrative, comp-based estimates, not projections validated with a model — built off what comparable companies charge and what similar audiences have shown willingness to pay for. Treat the ranges as a sense of shape, not a forecast.

Revenue stream	Pricing model	Basis for estimate	Illustrative range
Free map	No direct revenue — lead gen / SEO funnel	Data Center Map, Baxtel's public map	\$0 direct; drives everything below
Weekly brief	\$50–200/mo per seat	Comparable trade-intel newsletters	500–2,000 paid seats → ~\$0.3M–\$4.8M ARR
Screening tool	\$1,500–5,000/mo per org, enterprise	datacenterHawk / DC Byte quote-only enterprise pricing	50–150 org accounts → ~\$1M–\$9M ARR
Data licensing / API	\$25k–150k/yr per license	451 Research / S&P Global enterprise research pricing	20–60 licenses → ~\$0.5M–\$9M ARR
Financial-exposure marketplace take rate	0.5–2% of capital facilitated	Real-estate crowdfunding / RWA tokenization platform fee norms	\$50M–\$500M facilitated/yr → ~\$0.25M–\$10M
Token access fees (pay-per-query/report)	Micro-fee per query, paid in token	No direct comp — depends entirely on user/contributor volume	Unmodeled; small per-unit, scales with usage
Contribution/subscription rewards	Cost center, not revenue	Funded out of subscription + marketplace revenue above	Treat as customer-acquisition/data-acquisition cost

Potential Data Sources

Layer 1 — Global baseline, by geography

Geography	Sources	Access notes
Global baseline (all regions)	World Bank Open Data; IMF; ITU ICT indicators; GSMA Intelligence; UN Population Division	Covers every country at a basic level — start here everywhere, then layer in national statistics for precision
US & Canada	US Census Bureau + Bureau of Economic Analysis (regional GDP); Statistics Canada	Best granularity of any region — metro-area, even county-level detail available free
Europe & UK	Eurostat; national statistics offices (ONS UK, Destatis Germany, INSEE France)	Strong, standardized, comparable across countries thanks to Eurostat harmonization
Latin America	National statistics institutes (IBGE Brazil, INEGI Mexico); CEPAL/ECLAC regional data	Good national-level data; sub-national/metro detail varies a lot by country
Middle East	GCC-Stat; national statistics authorities (GASTAT Saudi Arabia, FCSA UAE)	Improving fast but still uneven — Gulf states publish more than others in the region
Africa	National statistics offices; Afristat (Francophone West/Central Africa); World Bank Africa Data Hub	The thinnest global coverage — exactly why this is also where contributor input matters most
Asia-Pacific	National Bureau of Statistics (China); MOSPI/Census (India); e-Stat (Japan); Australian Bureau of Statistics	Strong in developed markets (Japan, Australia, Singapore); China's data is extensive but should be cross-checked

Layer 2 — Data centers (existing & planned), by geography

Geography	Sources	Access notes
US & Canada	SEC EDGAR (public operators/REITs); state economic-development press releases; county planning/zoning portals; SEDAR+ (Canadian filings); trade press	Best-documented market in the world, but scattered across thousands of county sites — an aggregation problem, not an access problem
Europe & UK	Companies House (UK); national commercial registers (Handelsregister Germany, Infogreffe France); national investment-promotion agencies; DCD Europe coverage	Good corporate transparency; site-level planning data is more fragmented, country by country
Latin America	National investment-promotion agencies (ApexBrasil, Mexico's Ministry of Economy); local exchange filings (B3 Brazil, BMV Mexico); DCD LatAm coverage	Public-company disclosure is decent; private/JV deals need direct broker relationships
Middle East	Free-zone authorities (DMCC, ADGM); sovereign-wealth-fund press releases (PIF, Mubadala); Gulf exchange filings (Tadawul, DFM)	Mega-projects are heavily publicized; ownership structure behind them is often opaque and needs direct sourcing
Africa	Africa Data Centres Association; national ICT ministries; JSE (South Africa) filings; World Bank digital-infrastructure project database	South Africa and Kenya have real coverage; most of the rest of the continent is genuinely underserved
Asia-Pacific	China: SSE/SZSE filings + NDRC policy. Japan: EDINET + METI. India: Invest India + BSE/NSE. Singapore: EDB + SGX. Australia: ASX + state agencies	Wide range — Singapore and Australia are transparent; China and India require pairing government announcements with corporate filings

Layer 3 — Power grid & interconnection, by geography

Geography	Sources	Access notes
US & Canada	FERC eLibrary; public ISO/RTO interconnection queues (PJM, MISO, ERCOT, CAISO, SPP, NYISO); NERC; IESO (Ontario), Hydro-Québec	The gold standard for public interconnection-queue data — nowhere else in the world publishes this much detail
Europe & UK	ENTSO-E Transparency Platform; Capacitypedia (pan-European hosting-capacity platform, 27 countries); National Grid ESO/NESO (UK); RTE (France); German TSOs	Europe consolidated this in 2026 with Capacitypedia — worth building against directly
Latin America	ONS (Brazil); CENACE (Mexico); CAMMESA (Argentina)	Each country has one clear national operator — simpler than the US's multi-ISO landscape, but less granular queue detail
Middle East	National utilities (DEWA Dubai, Saudi Electricity Company, ADWEA Abu Dhabi)	Utility-controlled and far less transparent than the US or Europe — likely to lean on contributor/broker relationships here
Africa	National utilities (Eskom South Africa, Kenya Power, TCN Nigeria); African Development Bank grid-project database	Thin and inconsistent — the hardest layer to source here and the highest-value if we do

Asia-Pacific	State Grid Corporation of China / China Southern Power Grid; Grid-India + Central Electricity Authority; OCCTO (Japan); AEMO (Australia)	Australia and Japan publish well; China and India require government-portal monitoring rather than a single public queue
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Layer 4 — Power source & supporting infrastructure, by geography

Geography	Sources	Access notes
Global baseline	BloombergNEF (PPA and renewables tracking); corporate RE100/sustainability disclosures	Good starting point everywhere for the renewable-PPA piece specifically
US & Canada	State environmental agencies (air-quality permits); state water-rights agencies; FCC (fiber); connectivity directories	Permit-level detail is public but state-by-state — fifty-plus separate portals in the US alone
Europe & UK	National environmental agencies; EU Water Framework Directive reporting; national telecom regulators (Ofcom UK, Bundesnetzagentur Germany)	Water-use disclosure is becoming a bigger siting story in Europe than in the US
Latin America	National environmental agencies (IBAMA Brazil); national telecom regulators (Anatel Brazil, IFT Mexico)	Environmental permitting is public in principle, slow and paper-heavy in practice
Middle East	National environmental/water authorities; telecom regulators (TDRA UAE, CITC Saudi Arabia)	Water and cooling are the binding constraint in this region more than almost anywhere else
Africa	National environment/water ministries; national telecom regulators	Least aggregated data of any layer/region combination — largely a contributor-sourced layer here
Asia-Pacific	National environmental ministries; telecom regulators (TRAI India, Japan's MIC)	Coverage quality tracks overall institutional transparency — strong in Japan/Australia/Singapore, weaker elsewhere

Cross-cutting: ownership & corporate structure, by geography

Geography	Sources	Access notes
US & Canada	SEC EDGAR; state Secretary of State registries; UCC filings; SEDAR+ (Canada); PitchBook, Crunchbase	The easiest region to unwind an anonymous LLC, and still genuinely hard — this is where paid contributors add the most value
Europe & UK	Companies House (UK); Handelsregister (Germany); Infogreffe (France); EU Business Registers Interconnection System (BRIS)	BRIS is a real asset — one interconnection layer across most of the EU's national registries
Latin America	Junta Comercial / Receita Federal (Brazil); Registro Público de Comercio (Mexico)	National registries exist but aren't as interconnected as OpenCorporates or BRIS — manual cross-referencing needed
Middle East	DIFC/ADGM company registries; national commerce ministries	Free-zone entities are common and specifically designed for structuring privacy — expect this to be the hardest region for ownership unwinding
Africa	CIPC (South Africa); Corporate Affairs Commission (Nigeria)	Registries exist for the larger economies; coverage drops off fast

		elsewhere on the continent
Asia-Pacific	State Administration for Market Regulation (China); Ministry of Corporate Affairs (India); ACRA (Singapore); ASIC (Australia)	Singapore and Australia are clean and fast; China's registry is extensive but written for a different disclosure regime
Global overlay	OpenCorporates (aggregates many national registries); PitchBook, Crunchbase, Mergermarket for M&A history	The single best starting point before diving into any country's registry directly

Questions to Answer

- Which markets to cover first? Latin America? South Asia?
- Which data is accessible easily but not already picked over and displayed on other sites?
- What is a way to reward users for submitting useful data and how do we verify if it is useful?
- What are some ways we can display the data in “layers” so that a person can see data centers and then the downstream infrastructure that is required and that will result from its construction — since all are investable?
- How can we obscure the data in a clever way before the paywall is accessed?
- What are some ways we can build an MVP of this map, similar in spirit to a narrow, single-dataset MVP map?